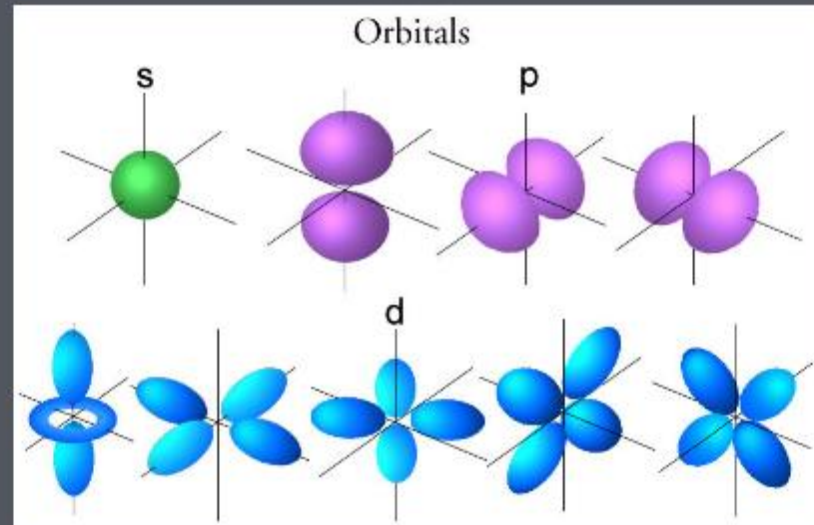


Biophysics (Bio361)

Dr. Arjun Ray
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Interactions - it's all electrons

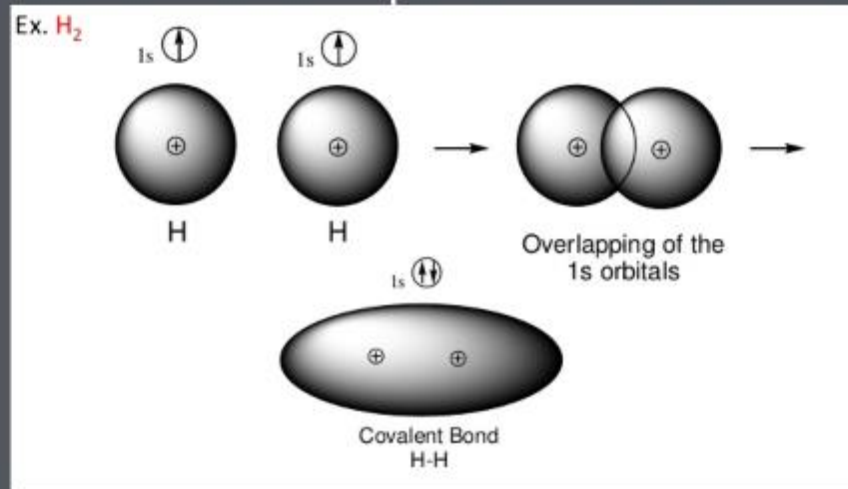


The Pauli exclusion principle (QM): Two identical electrons cannot occupy the same quantum state simultaneously

Once a particular orbital has one electron with spin "up" and another with spin "down", it is full

This explains why atoms repel very strongly at short distance

Orbitals explain bonds



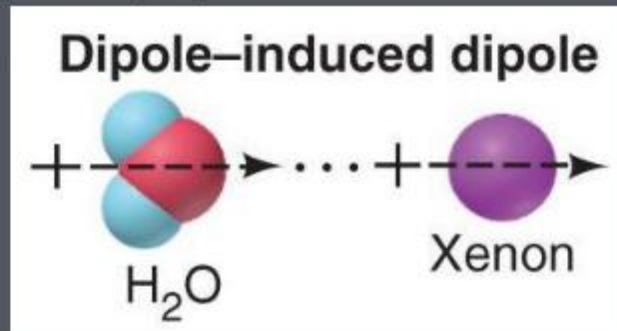
Two individual atoms each have one electron with spin “up” in the first (S) orbital

By combining these and delocalizing them, we can have a filled orbital (which is advantageous) with spin up & down

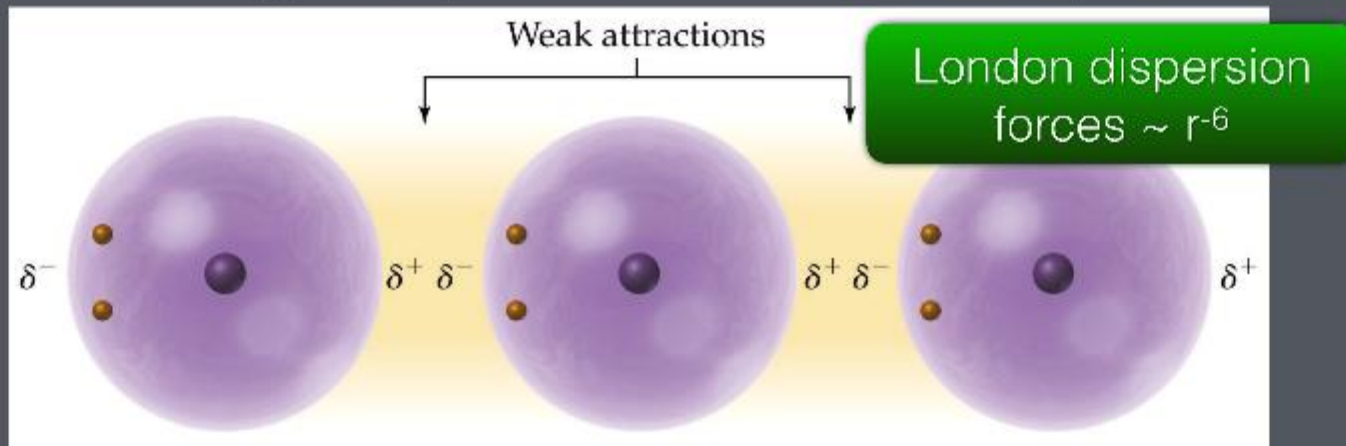
This is how covalent/hard bonds are formed

Conversely, saturated orbitals will repel each other

What happens at large r ?

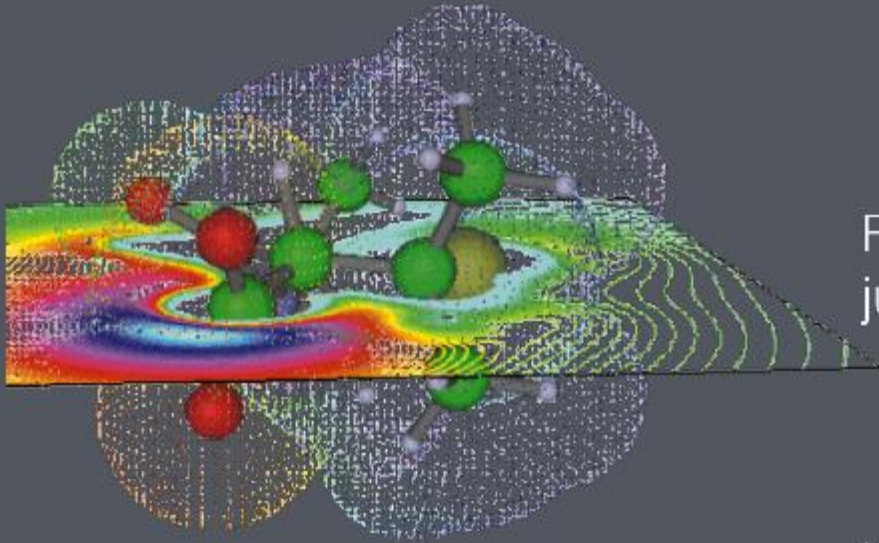


A charge or dipole can *induce* another dipole



Spontaneous (thermal) fluctuations will induce dipoles

Do we need QM (here)?



Friend of order: Yes, because we just spend three slides justifying it!

- We can only handle ~ 100 atoms
- To be really accurate, you should use time-dependent relativistic QM (think... 1 electron)
- No atomic motions $\Rightarrow$ 0K
- No water? Vacuum, seriously?

In principle we could model a complex system by starting
from quantum chemistry to derive all parameters
PROBLEM: Extrapolating 15 orders of magnitude will kill you

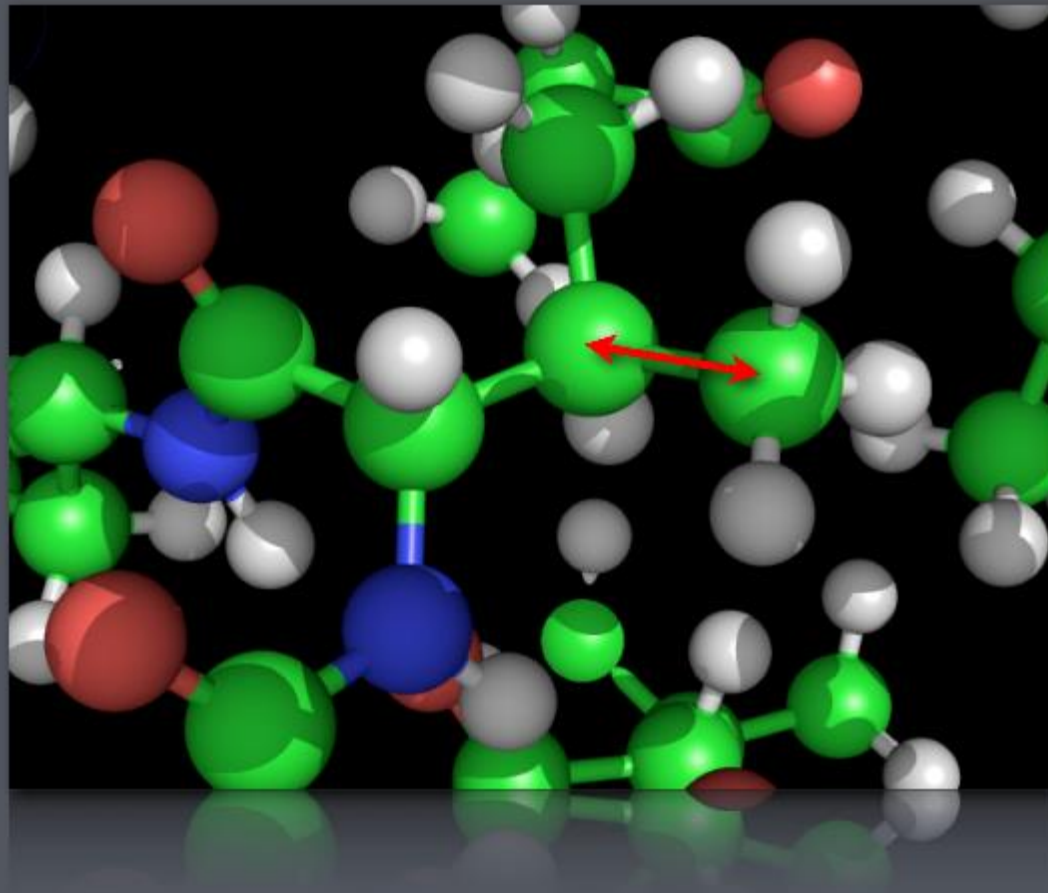
Shneior Lifson, Arie Warshel, Mike Levitt (1960s):
Semi-empiric models of biomolecules
Use simple interactions that you fit to experiments

Made it possible to simulate protein motion
and explain the function of enzymes



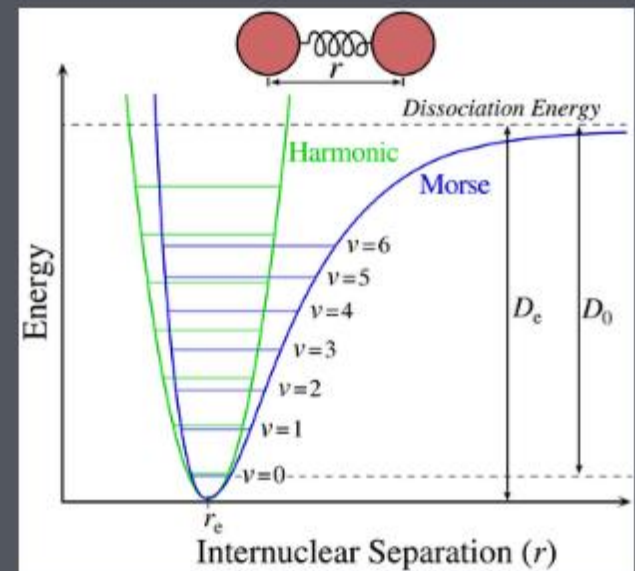
Nobel Prize Chemistry 2013

Bond stretching

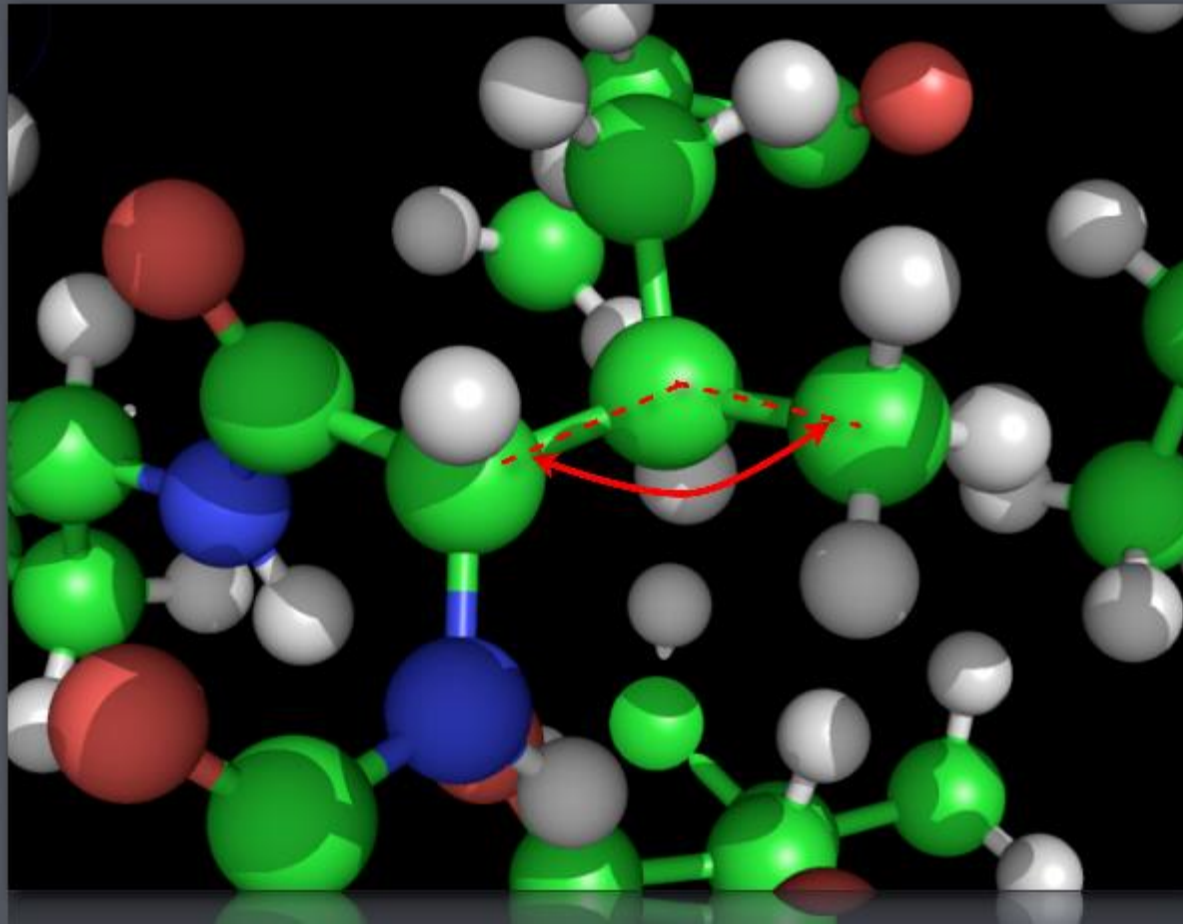


$$V = k \Delta x^2$$

$$V = D (1 - e^{-ax})^2$$



Angle vibrations

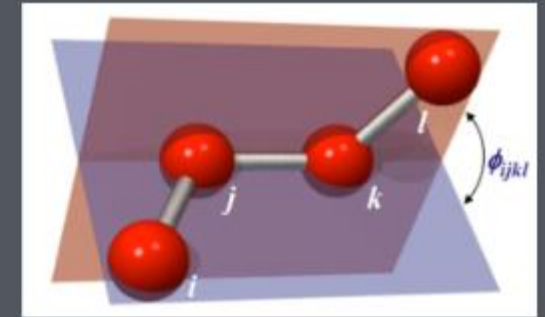
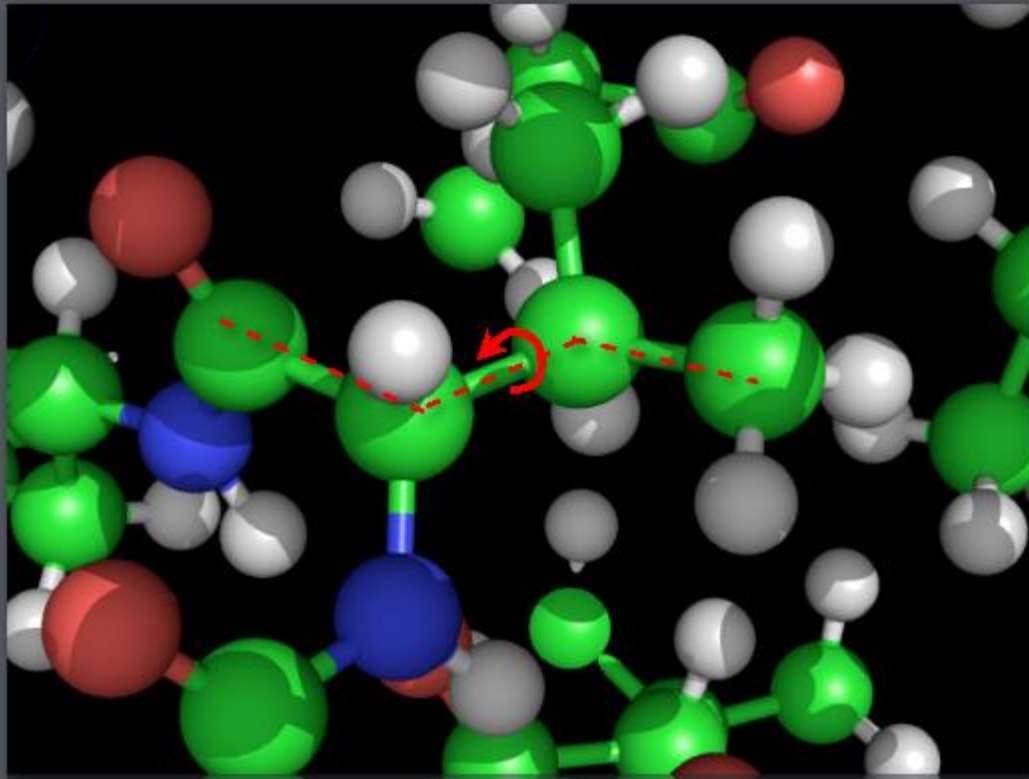


Not quite as rigid as bonds at 300K, but almost.

Angles move a couple of degrees.

Similar to bonds:
Should really be a QM oscillator, but works great with classical approximation

Torsion angles

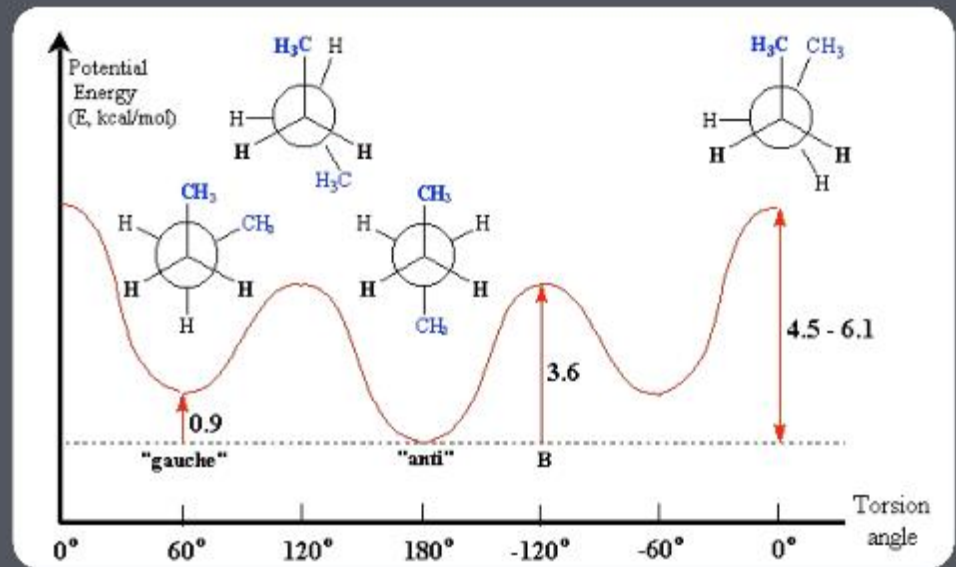
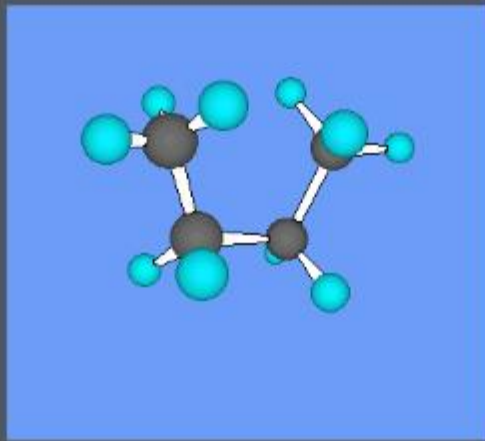
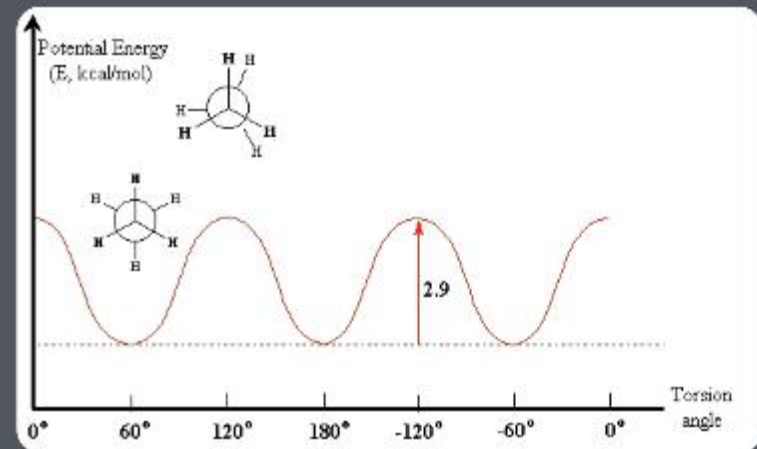


Torsion=dihedral angle

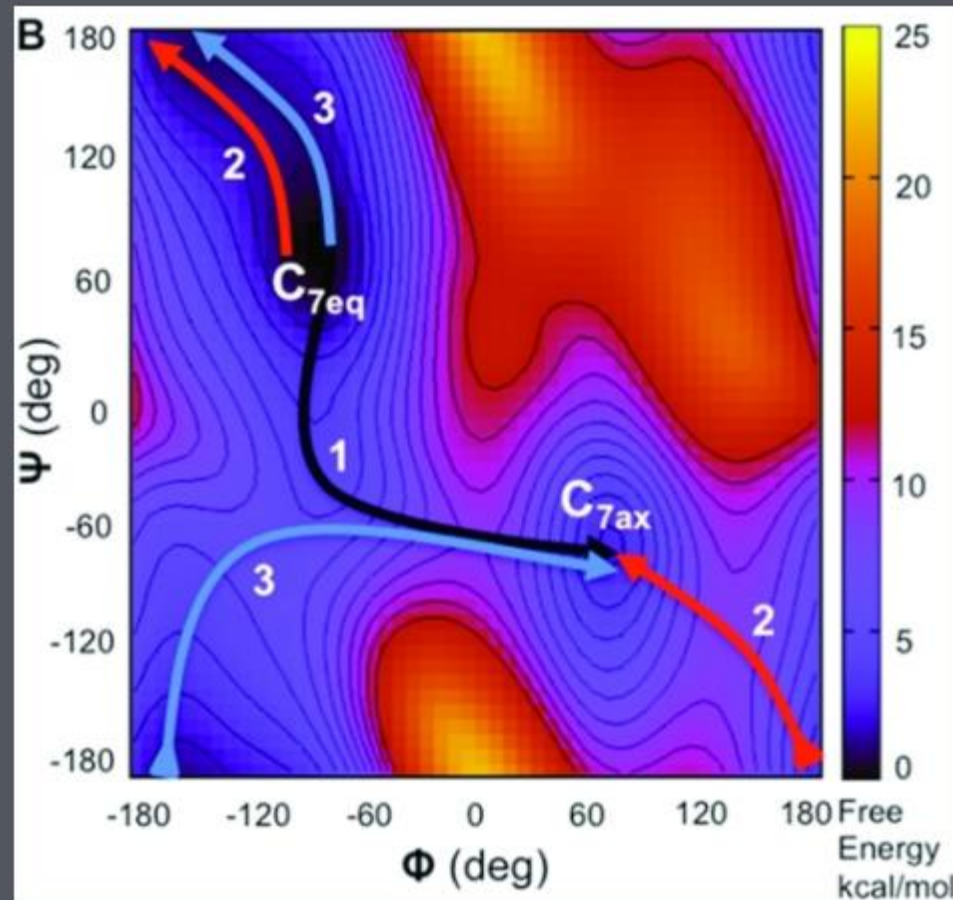
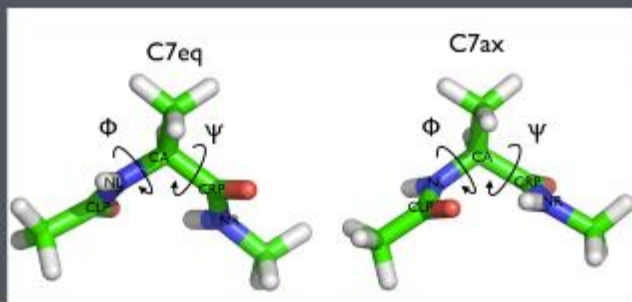
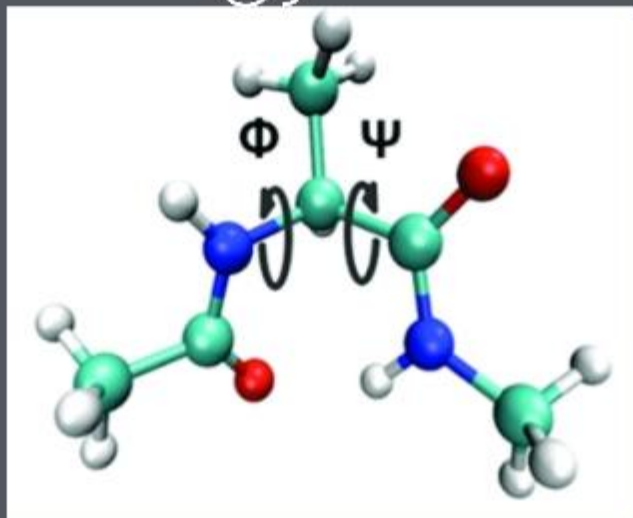
Angle between planes
formed by atoms i-j-k
and atoms j-k-l

Gives rise to the degrees of freedom in the Ramachandran diagram
Dihedrals are easy to alter!

A few torsion potentials

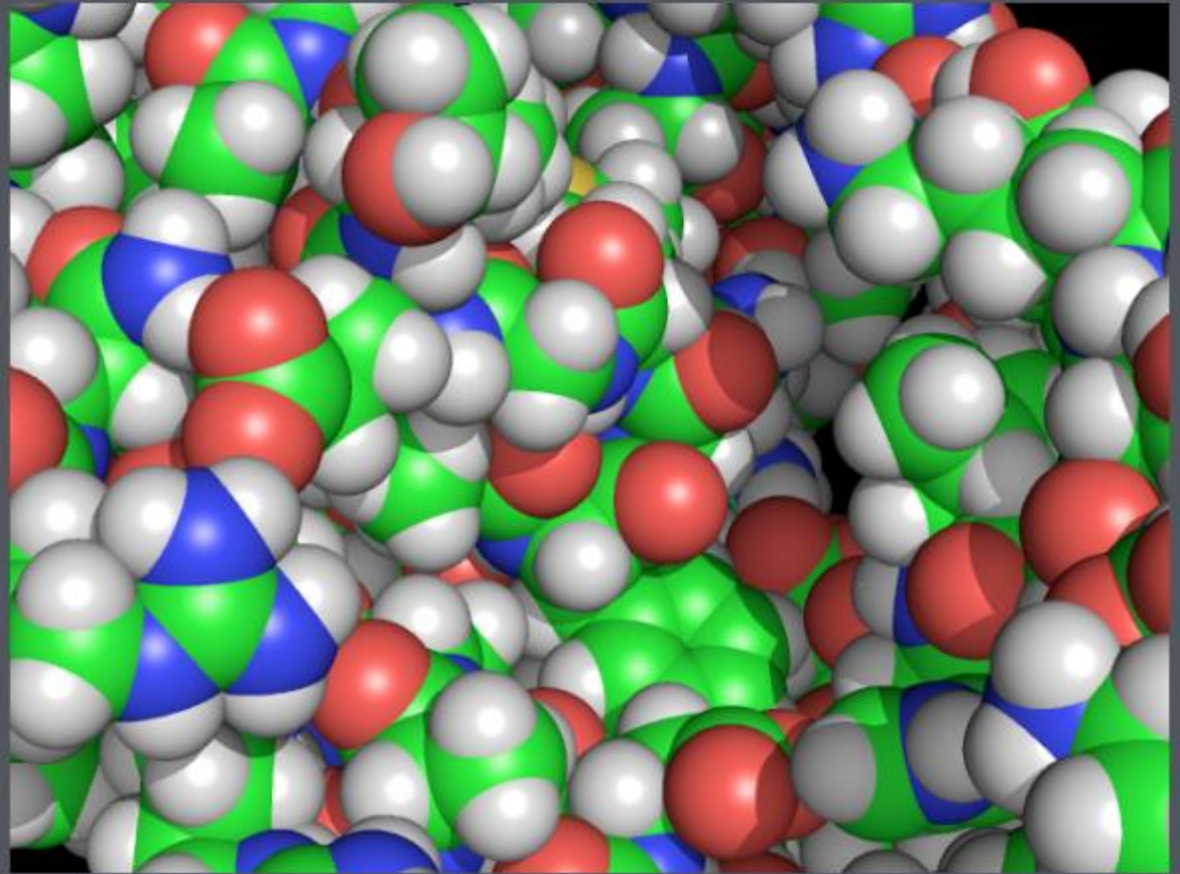


Energy as function of 2 torsions

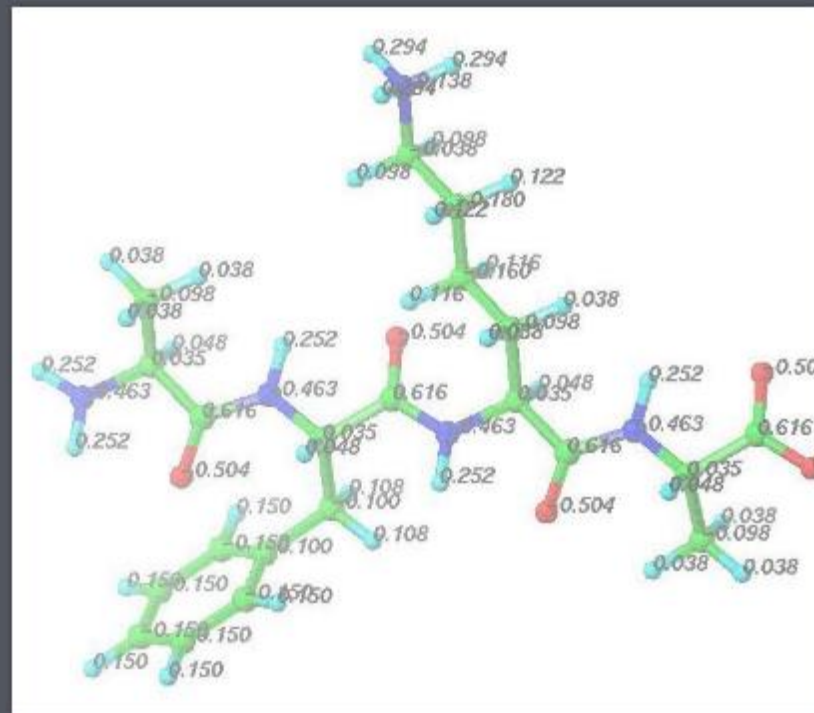
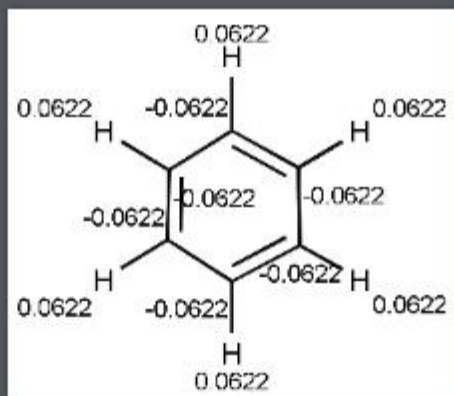
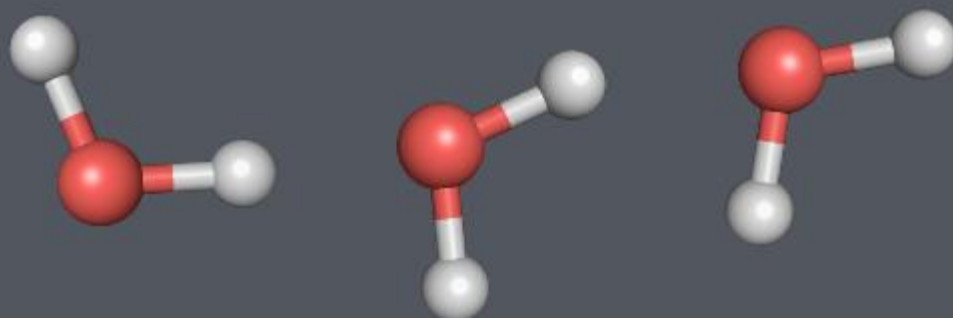


Non-bonded effects

- Charge interactions
Non-bonded
effects(electrostatics)
- Packing effects
(van der Waals
interactions)



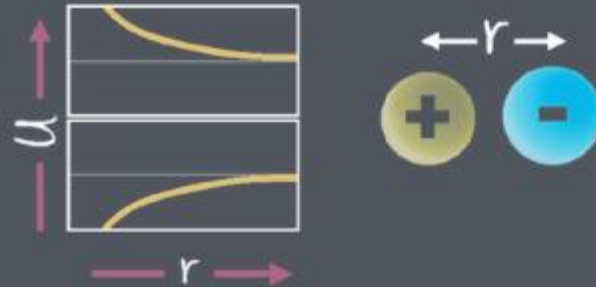
Partial Charges



Electron clouds are mobile in molecules, with density varying between different atoms!

Electrostatic strength

$$+ \sum_{\text{All partial charges}} 332 q_i q_j / r$$



Electrostatic interactions decay as $1/r$ (slow!)

Example interaction energy:

Two charges separated by $\sim 1 \text{ \AA}$: 330 kcal/mol!
(Compare to bond rotation, 2-4 kcal)

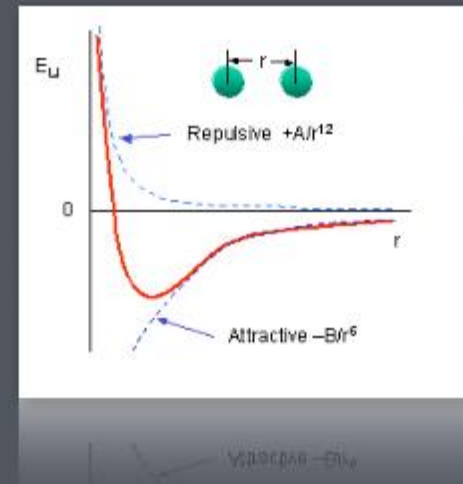
Van der Waals interactions

Pauli: Atoms repel each other at close distance due to overlap of electrons (strong repulsion)

London: All atoms attract each other at long distance due to induced dipole effects (weak dispersion)

Example - Buckingham potential:

$$V(r) = A \exp^{-Br} + \frac{C_6}{r^6}$$



exp(r) is expensive (slow) to calculate on most computers

Lennard-Jones interactions

Simpler form than Buckingham

In practice, atoms should never approach really close, so we just want a basic model of the repulsion

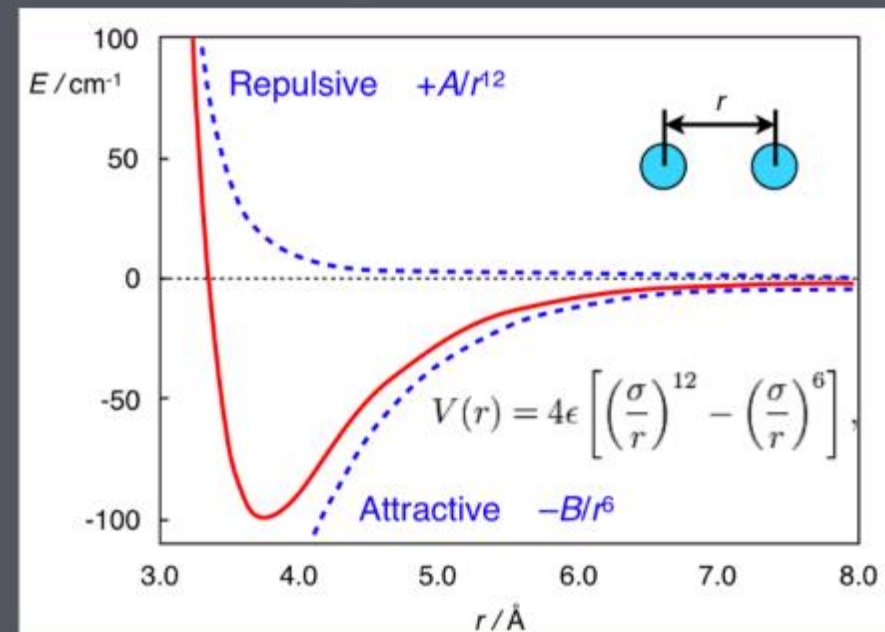
Smart trick: When we have calculated $1/r^6$, it is trivial to get $1/r^{12}$ (1 multiplication)

Lennard-Jones potential

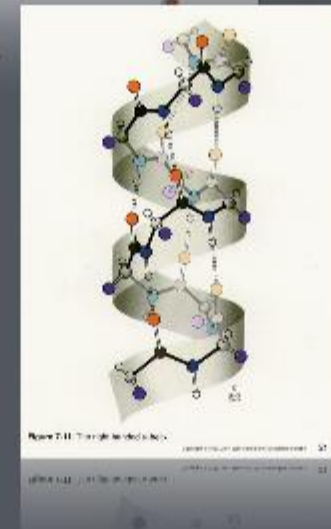
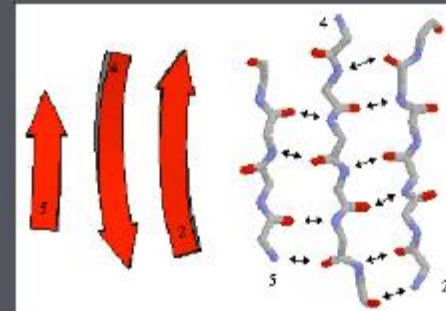
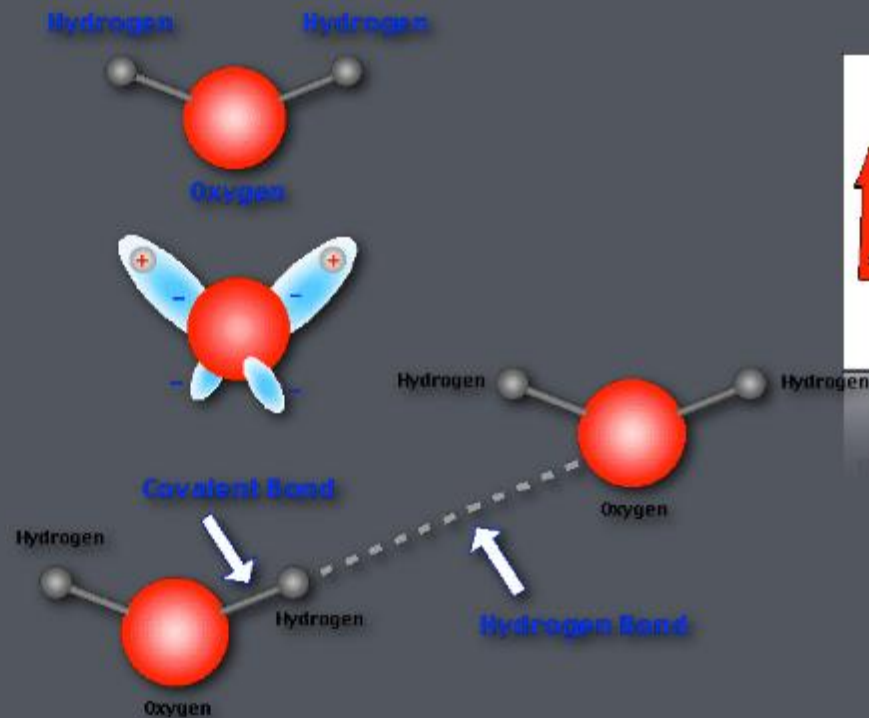
$$V(r) = \sum_{i=1}^N \sum_{j=1}^N \left(\frac{C_{12}}{r_{ij}^{12}} - \frac{C_6}{r_{ij}^6} \right)$$

Table 3.1. Typical parameters of van der Waals interaction potentials.

Interaction	E_0 (kcal mol ⁻¹)	r_0 (Å)	$r_{\min}$ (Å)	Minimum van der Waals radius of atom (Å)	
H...H	0.12	2.4	2.0	H:	1.0
H...C	0.11	2.9	2.4		
C...C	0.12	3.4	3.0	C:	1.5
O...O	0.23	3.0	2.7	O:	1.35
N...N	0.20	3.1	2.7	N:	1.35
CH ₂ ...CH ₂	≈0.5	≈4.0	≈3.0	CH ₂ :	≈1.5

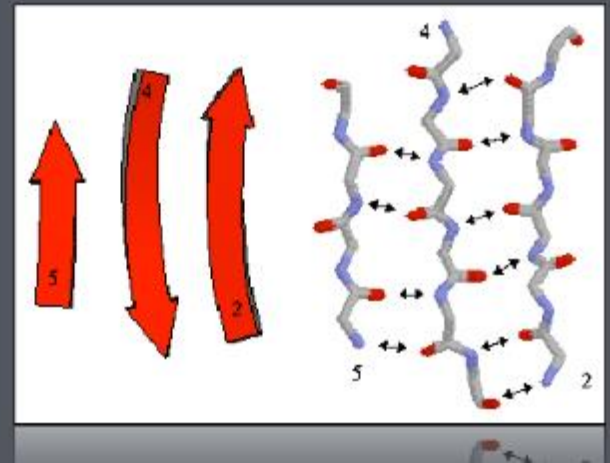
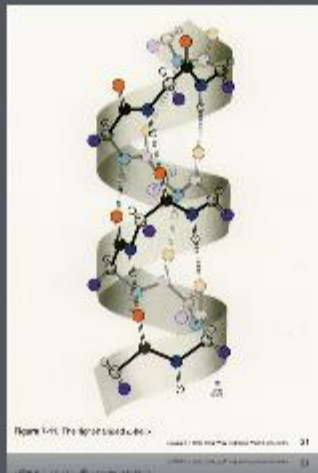
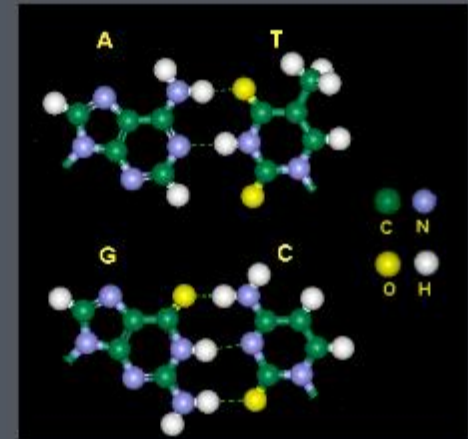
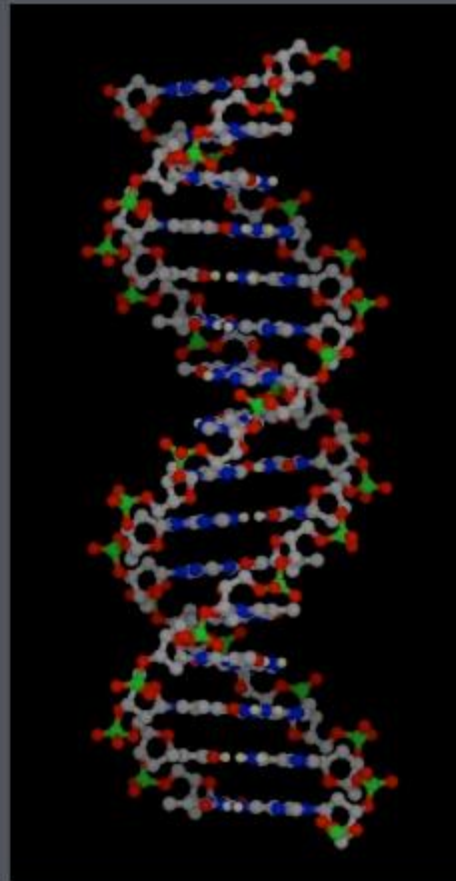
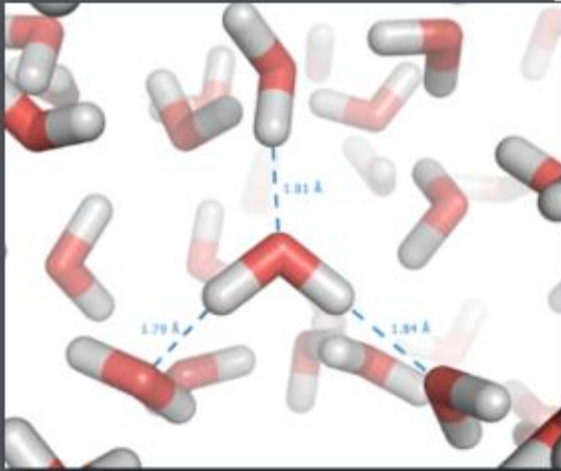


Hydrogen bonds in proteins

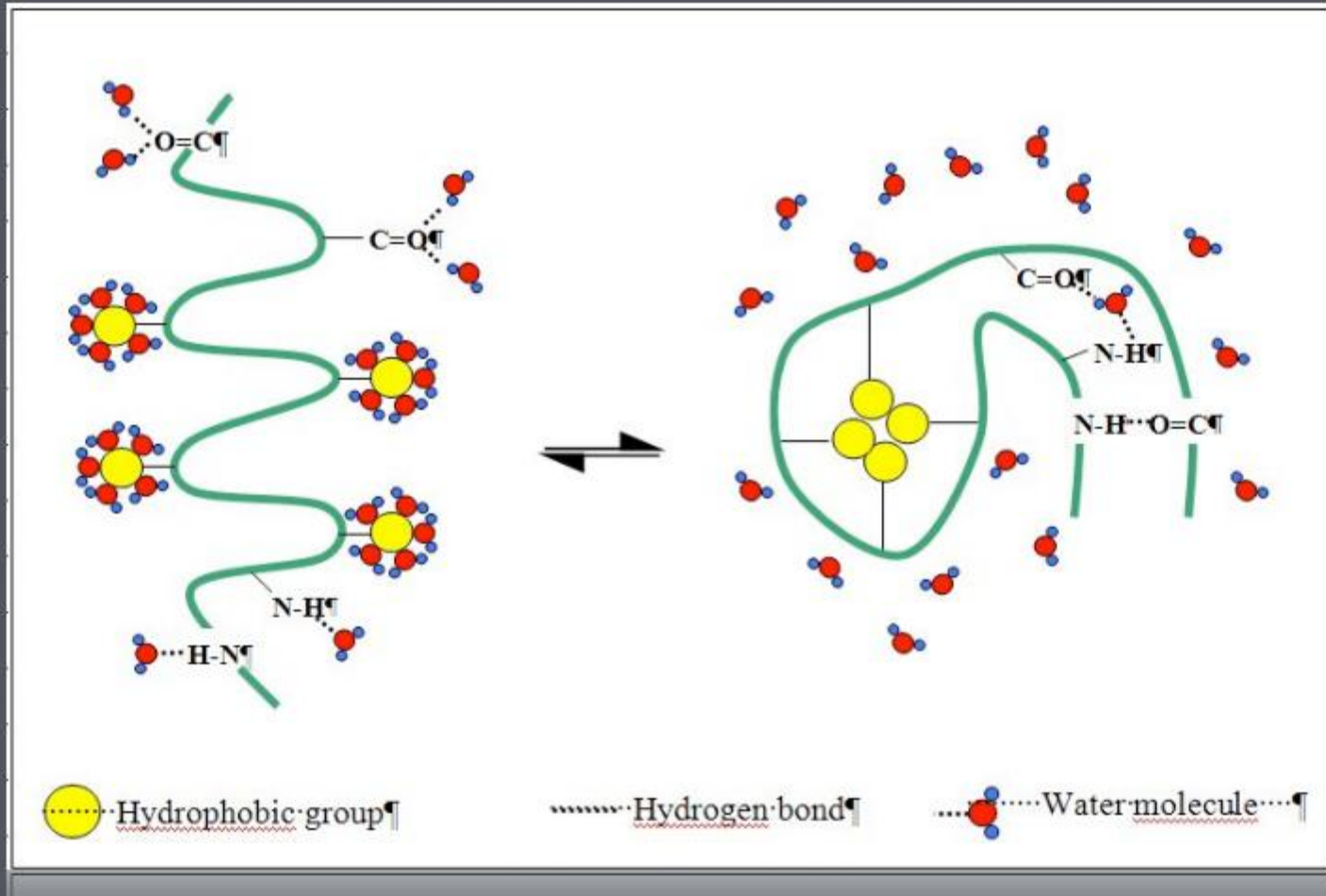


Lots of energy: Waters will try to move any way they can to maintain H-bonds

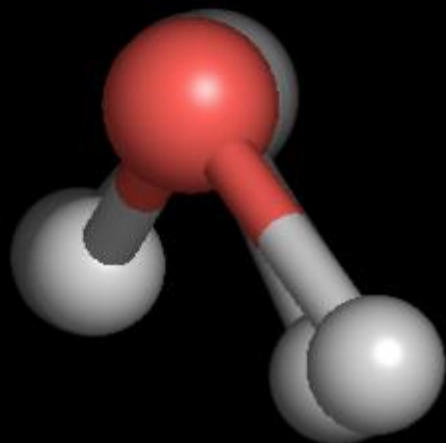
Hydrogen bonds



Hydrogen bonds & Protein folding

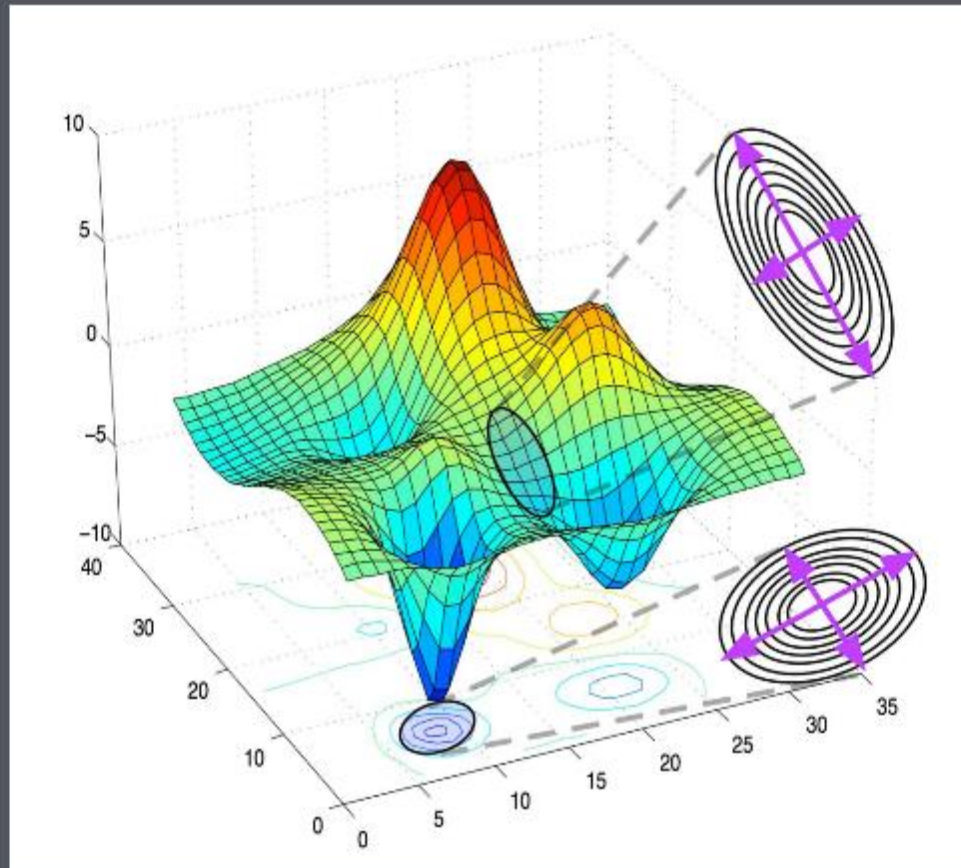


8 fs



$$\begin{aligned}
 V(r) = & \sum_{bonds} \frac{1}{2} k_{ij}^b (r_{ij} - r_{ij}^0)^2 \\
 & + \sum_{angles} \frac{1}{2} k_{ijk}^\theta (\theta_{ijk} - \theta_{ijk}^0)^2 \\
 & + \sum_{torsions} \left\{ \sum_n k_\theta [1 + \cos(n\phi - \phi_0)] \right\} \\
 & + \sum_{impropers} k_\xi (\xi_{ijkl} - \xi_{ijkl}^0) \\
 & + \sum_{i,j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \\
 & + \sum_{i,j} \left[\frac{C_{12}}{r_{ij}^{12}} - \frac{C_6}{r_{ij}^6} \right]
 \end{aligned}$$

A high-dimensional energy landscape



Covered material and suggested reading

- Finkelstein
 - Chapter 3 & 4
- David Eisenberg - "The discovery of helices and sheets, the principal structural features of proteins", PNAS, 2003
- Cyrus Levinthal - "Molecular Model building by Computer", Scientific American, 1966

Study questions

1. Name three properties by which amino acids can differ.
2. One amino acid is NOT chiral. Which one? Is it special in any way?
3. There is one more highly special amino acid. Which one? How is it special?
4. What would happen if we had some D-isomer amino acids in a protein?
5. How are amino acids linked into a protein chain?
6. What are the levels of structure organization in proteins?
7. Describe the relation between sequence, structure, and function.
8. Does function induce structure?
9. Which ones are the most important degrees of freedom in proteins?
10. Why are cis conformations almost never observed in peptide bonds?
11. What did (1) Anfinsen and (2) Levinthal say?
12. Explain what a Ramachandran plot can tell you about protein geometry.
13. What is the major difference in formation of helices and sheets.
14. Why do chemical bonds form?
15. What are the interactions related to bonds between atoms in a protein?
16. What are the non-bonded interactions in a protein?
17. What interactions are strongest? Weakest?
18. What interactions do you have most of?
19. Compare electrostatics with dispersion. Strength?